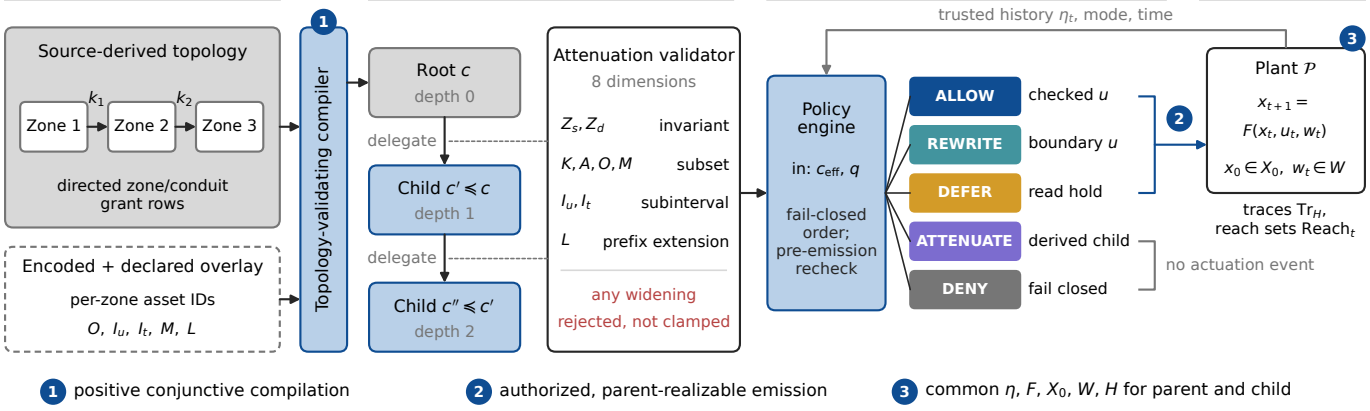
Policy input
Capability lineage
Runtime authorization
Plant
Source-derived topology
Zone 1
Zone 2
Zone 3
k_1
k_2
directed zone/conduit grant rows
Encoded + declared overlay
per-zone asset IDs
O, I_u, I_t, M, L
1
Topology-validating compiler
Root c
depth 0
delegate
Child $c' \preccurlyeq c$
depth 1
delegate
Child $c'' \preccurlyeq c'$
depth 2
Attenuation validator
8 dimensions
Z_s, Z_d invariant
K, A, O, M subset
I_u, I_t subinterval
L prefix extension
any widening rejected, not clamped
trusted history η_t, mode, time
Policy engine
in: c_{eff}, q
fail-closed order; pre-emission recheck
ALLOW
checked u
REWRITE
boundary u
DEFER
read hold
ATTENUATE
derived child
DENY
fail closed
2
no actuation event
3
Plant $\mathcal{P}$
$x_{t+1} = F(x_t, u_t, w_t)$
$x_0 \in X_0,\ w_t \in W$
traces Tr_H, reach sets Reach_t
1 positive conjunctive compilation
2 authorized, parent-realizable emission
3 common η, F, X_0, W, H for parent and child